Pranjal Raj

Software Engineer & MERN Stack Developer

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Technical Skills

- **Programming Languages:** Java, C/C++, JavaScript, Typescript, R, Dart
- **Development:** HTML5, CSS, Javascript, Typescript, React
- **Development Workflow:** Git, Github, VS Code
- **Backend:** Nest.js, Node.js, Express.js, RESTful APIs, WebSockets, Firebase
- **DevOps :** Docker, Docker Compose, GitHub Actions, Amazon Web Services (AWS), Google Firebase, Redis, Kafka, DynamoDB
- **Monitoring :** Prometheus, Grafana

Experience

Web Developer Intern Hybrid
Emerald Windows October 2026 – April 2027

- Developing responsive, client-facing web applications using React.js, Next.js, TypeScript, JavaScript, and Tailwind CSS, focusing on usability, reusable components, and maintainable code.
- Building interactive user interfaces using Vite, Framer Motion, Lucide React, and shadcn/ui to deliver consistent and engaging user experiences.
- Contributing to full-stack development by implementing backend functionality, RESTful APIs, data validation with Zod, and server-side logic using NestJS and JavaScript.

Mobile Systems Engineering Intern Remote
Ardent Software Services June 2026 – August 2026

- Built high-performance Flutter mobile app with real-time data synchronization, achieving 60 FPS and <500ms sync latency through asynchronous APIs, SQLite caching, and race condition prevention.
- Optimized concurrent state management using Dart isolates for background processing, reducing memory footprint by 40% and maintaining 99.8% app stability across test deployments.
- Implemented graceful degradation patterns for offline-first architecture, ensuring zero data loss during network transitions and reliable synchronization with cloud backend.

Projects

Panchayat App (Society Mobile Application) GitHub: Project Link
React 18+, TypeScript, Tailwind CSS, Lucide React, Vite, Node.js, Express.js, Secure API Layer

- Developed a full-stack cooperative housing management platform using React 18+, TypeScript, and Node.js, centralizing administrative and resident communication into a unified digital ecosystem.
- Devised role-based authentication, secure profile management, and integrated subscription/payment flows, directly addressing chronic inefficiencies in record-keeping, transparency, and fee collection for housing societies.
- Designed a mobile-first, responsive UI with Tailwind CSS and Framer Motion micro-animations, delivering a polished, high-contrast user experience across all device types.
- Engineered a secure server-side API layer with Express.js to handle authentication, data management, and critical operations, ensuring sensitive data remains protected and never exposed client-side.

IntelliBin (Software + Hardware) GitHub: Project Link
Google Firebase, HTML5, CSS, JavaScript, Node.js, Real-time Web Dashboard

- Designed complete IoT system architecture for intelligent waste management, integrating hardware sensors (ultrasonic, gas, temperature, moisture, GPS) with cloud infrastructure for real-time monitoring and optimization.
- Built embedded system firmware on ESP32 using Arduino IDE with asynchronous sensor data collection, handling concurrent I/O operations and synchronization of multi-sensor inputs at real-time intervals.
- Engineered data aggregation pipeline using HTTP APIs to transmit sensor data to Firebase with custom JSON payload optimization, reducing bandwidth usage by 45% through compression and selective field transmission.
- Engineered intelligent alert system with threshold-based triggers for hazardous gas detection and bin overflow prediction, optimizing collection routes based on spatial data for 25% reduction in collection time.

Education

Vellore Institute of Technology Vellore, Tamil Nadu
B.Tech in Electronics and Communication (CGPA: 8.30) Aug 2024 – 2028

Relevant Coursework: Operating Systems, Database Management Systems, Computer Networks, Data Structures & Algorithms, Software Engineering, Microprocessors & Interfacing.

Mount Litera Zee School, Roorkee – *Grade XII, PCM with Computer Science* (Perc.: 85%) July 2022 – May 2023
Mount Litera Zee School, Roorkee – *Grade X* (Perc.: 96%) July 2021

Co-Curriculars

Clubs and Chapter Participation Jul 2024 – Present
VIT, Vellore, Tamil Nadu

- Engineered the landing page for HackOIT V4.0 and managed sponsorship communications, contributing to a 20% increase in total event registrations.
- Led PR and media outreach for community initiatives, increasing event participation by 30% through targeted social media campaigns and strategic content creation.

Certifications

- **DSA with Java** (Delta Batch) – Apna College.
- **MERN Fullstack Program** – Apna College.
- **Amazon Web Services Program** – Udemy.
- **Software Engineering Job Simulation** – Commonwealth Bank.